

Research Statements

The research statements afford you the opportunity to articulate your academic plans in relation to the objectives of the DOE CSGF. Reviewed in the context of your complete application, your responses shed important light into your way of thinking as influenced by your experiences and goals.

Field of Interest

- a. *In terms a general audience would understand, describe an important, outstanding challenge in mathematics, statistics or computer science that you would like to pursue in your research. (1/3)*
- b. *Describe the particular mathematics, statistics or computer science problem that you would like to pursue in your research. What would be the impact on high-performance computing and on science, engineering and/or society in general if this challenge could be successfully addressed? (2/3)*

Modern machine learning, dominated by deep neural networks, has become a surprisingly empirical field. The opacity of typical deep learning models makes improving an existing model difficult and interpreting its behavior almost impossible. As a result, trial-and-error forms a major part of modern machine learning model design, and many improvements are justified using general principles or post-hoc rationalization that cannot be justified theoretically. I am interested in bridging the gap between provable, theoretical results and the empirical effectiveness of modern machine learning techniques. During my undergraduate studies at Duke, I have worked on research across the spectrum of low-level numerical methods and high-level deep learning applications -- now, I am interested in unifying these two lines of work, applying recent innovations in machine learning to challenges in scientific computing and engineering.

I am particularly interested in solving inverse problems using machine learning. I have briefly encountered this area of research from two contrasting perspectives in my studies so far. At Lawrence Livermore National Laboratory, I developed numerical methods for the interface reconstruction problem, a key part of volume-of-fluids simulation techniques where the fluid boundaries must be predicted from fluid volume information. At Duke, I have worked on machine learning techniques for face super-resolution, where the goal is to create a high-resolution version of a given blurry image. The key technique I explored in this project is the use of pretrained deep generative models to act as a powerful prior for complex image generation tasks [1]. Recent work [2, 3] has shown that this concept can be extended to solve challenges ranging from denoising to MRI data collection. Furthermore, using arguments from the compressed sensing literature, these methods can be shown to be provably robust. Further insights into deep generative priors could lead to more efficient techniques for inverse problems throughout materials science, physics simulations, and medical imaging.

Citations (Optional)

1. Menon et al. PULSE: Self-Supervised Photo Upsampling via Latent Space Exploration of Generative Models. CVPR, 2020.
2. Bora et al. Compressed sensing using generative models. ICLR, 2017.
3. Jalal et al. Robust compressed sensing MRI with deep generative priors. NeurIPS, 2021.

High-Performance Computing

- a. *What is the most complex calculation you have run on a high-performance machine as part of your research experience? Or if you haven't run a high-performance computing system, tell us about the most complex computational problem you have tackled. (1/2)*
- b. *Imagine if you were given access to resources 100 times more powerful than what you have access to. What would that enable you to do, and what do you perceive the mathematical and computer science challenges to be? (1/2)*

Over the past two summers, I have participated in the High-Energy Density Physics program at Lawrence Livermore National Laboratory, working on low-level geometric/numerical methods for computational fluid dynamics. My project began with the insight that as massively parallel systems become more prevalent, higher-order methods (computationally costly but parallelizable) may prove more useful than faster but less accurate lower-order techniques. Here, I tackled the interface reconstruction problem, where the goal is to track the changing shape of fluid boundaries as the materials move, from a unique perspective based on geometric inverse problems. Over my summers, I implemented a toolkit of algorithms for efficiently computing geometric regions of intersection and designed optimization algorithms for solving the local reconstruction problems; I also independently developed a separate marker particle for whenever the inverse problems are ill-defined. My final hybrid method allows for accurate tracking of fluid boundaries with far coarser meshes than preexisting interface reconstruction methods.

More recently, as I am working toward publishing my work, we have been interested in evaluating my algorithms on larger-scale physics simulations on the lab's supercomputer systems. Whereas my techniques run on a laptop might be able to solve a basic toy problem with a couple hundred elements, the lab's computing resources, like the flagship supercomputer Sierra, would be able to accommodate huge simulations for fusion reactions, drug design, or plasma physics. Although my current algorithms are designed to exploit parallelism, they still require the solution of small non-convex optimization systems using iterative methods. I believe this presents a key challenge and opportunity in the coming years. Recent works have shown the promise of applying deep neural networks to challenges traditionally solved using numerical or iterative techniques, including in the field of interface reconstruction [1, 2], but more work needs to be done to ensure that machine learning systems for scientific computing are implemented in a principled, robust way.

Citations (Optional)

1. Ataei et al, 2020. NPLIC: A Machine Learning Approach to Piecewise Linear Interface Construction.
2. Fenn et al. Material Interface Reconstruction Using Machine Learning. The International Conference for High Performance Computing, Networking, Storage, and Analysis, 2020.

Program of Study

Describe how the courses listed in your planned program of study would help prepare you to address the challenges you have described in questions 1 and 2. Discuss your rationale for choosing these courses. How will the science or engineering application courses you have selected impact your research?

Throughout my studies so far, I have sought to be a generalist as much as possible. I have developed a strong background in pure math and computer science theory through coursework, while doing research in topics including natural language processing, computer vision, and computational fluid dynamics. Propelled by constant exploration, I believe I have prepared the necessary background for rigorous interdisciplinary work, bridging the gap between principled numerical techniques and modern black-box deep learning.

Despite my training in pure math, I am fundamentally invested in the applications of this direction of work -- the development of numerical methods has been constantly shaped by the demands of the scientific applications, not just in the mathematical details of each problem but the very types of approaches that are even considered. I believe that it is crucial for me to develop a deep understanding of the entire scientific pipeline, from the fundamental physical/engineering demands to classical methods to modern approaches. I am particularly interested in applications to fluid dynamics and materials science -- these are areas that are relevant from a scientific/societal perspective and have already shown promise with machine learning techniques. To better design principled algorithms however, I also need to understand the pure math behind existing classical techniques, such as numerical PDEs, numerical linear algebra and functional analysis. The machine learning classes I have chosen represent some of the strongest provable algorithms currently in the field, which is crucial for designing robust systems. Finally, an understand of high-performance computing system is a must, since every method must consider the details of implementation.

Program of Study - Course List

Course number	Course Title
<i>Science & Engineering Applications</i>	
APPPHYS229	Statistical Mechanics of Learning and Computation
BIOE 214	Representations and Algorithms for Computational Molecular Biology
ME 351A	Fluid Mechanics
<i>Mathematics and Statistics</i>	
CME 307	Optimization
MATH 226	Numerical Solution of Partial Differential Equations
MATH 228	Stochastic Methods in Engineering
<i>High-Performance Computing</i>	
CME 213	Introduction to parallel computing using MPI, openMP, and CUDA
<i>Computer Science and Computer Engineering</i>	
CS 228	Probabilistic Graphical Models
CS 229S	Systems for Machine Learning
CS 234	Reinforcement Learning